

# IKERUSIKI Theory

A Boundary Coordinate Framework for  
Structural Reconstruction of Integers

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*Sequential Phase Orchestration*

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## Abstract

IKERUSIKI Theory proposes a structural framework for representing integers through Boundary Coordinates rather than exhaustive numerical enumeration. The mathematical framework was established through Verification001–030 and subsequently evaluated through four computational applications. Application001 demonstrated structural reconstruction for RSA2048-oriented composite integers. Application002 extended the framework to large integers up to 4096 bits. Application003 introduced computational optimization while preserving the mathematical framework. Application004 benchmarked the optimized implementation against the baseline implementation. These results indicate that the proposed framework can reconstruct structural candidates while maintaining a fixed mathematical formulation.

**Keywords:** Boundary Coordinate; Candidate Reconstruction; Structural Modulus; Prime Coordinate; Computational Number Theory; RSA2048

## Notation

| Symbol         | Definition                                  |
|----------------|---------------------------------------------|
| $B$            | Finite radix set, $B = \{2, 3, \dots, 26\}$ |
| $b$            | Radix                                       |
| $n$            | Integer under consideration                 |
| $r_b(n)$       | Residue $(n - 1) \bmod b$                   |
| $D_b(n)$       | Boundary Distance                           |
| $BC_B(n)$      | Finite Boundary Coordinate                  |
| $BC_\infty(n)$ | Infinite Boundary Coordinate                |
| $Cand_B(n)$    | Candidate Reconstruction Set                |
| $\Phi_B(n)$    | Core Mathematical Equation                  |
| $\mathbb{P}$   | Set of prime numbers                        |
| LIMIT          | Computational verification range            |

## 1. Introduction

Traditional computational number theory primarily represents integers by their numerical values. IKERUSIKI Theory instead represents integers through structural information derived from Boundary Coordinates.

The mathematical framework was established through Verification001–030, introducing Boundary Distance, Boundary Coordinates, Candidate Reconstruction, and the Core Mathematical Equation. Following the completion of the theoretical framework, four computational applications were conducted: Application001 for RSA2048-oriented composite integers, Application002 for large integers up to 4096 bits, Application003 for computational optimization, and Application004 for benchmark analysis.

The purpose of this paper is to present the completed mathematical framework, summarize its computational verification, and provide a reproducible structural representation of integers based on Boundary Coordinates.

## 2. Mathematical Framework

### 2.1 Structural Observation

IKERUSIKI Theory represents integers by structural information derived from modular boundary relationships rather than by numerical value alone. For each radix  $b$ , an integer is represented by its Boundary Distance.

#### Definition 1 (Boundary Distance)

For an integer  $n$  and radix  $b$ , define

$$r\_b(n) = ((n - 1) \bmod b).$$

The Boundary Distance is defined as

$$D\_b(n) = \min(r\_b(n), b - 1 - r\_b(n)).$$

#### Definition 2 (Finite Boundary Coordinate)

Let

$$B = \{2, 3, \dots, 26\}.$$

The finite Boundary Coordinate is defined as

$$BC\_B(n) = \{D\_b(n) \mid b \in B\}.$$

#### Definition 3 (Infinite Boundary Coordinate)

The Infinite Boundary Coordinate extends the radix family to every integer satisfying  $b \geq 2$ :

$$BC\_∞(n) = \{D\_b(n) \mid b \geq 2\}.$$

#### Definition 4 (Candidate Reconstruction)

For a finite radix family  $B$ , define the candidate reconstruction set by

$$Cand\_B(n) = \{x \in \mathbb{N} \mid \forall b \in B, (x - 1) \bmod b \in \{D\_b(n), b - 1 - D\_b(n)\}\}.$$

#### Definition 5 (Core Mathematical Equation)

The structural reconstruction is obtained by

$$\Phi\_B(n) = Cand\_B(n) \cap [1, LIMIT] \cap \mathbb{P},$$

where  $\mathbb{P}$  denotes the set of prime numbers.

### 3. Core Mathematical Equation

The Core Mathematical Equation provides the mathematical foundation of IKERUSIKI Theory. Rather than identifying an integer directly by its numerical value, the framework reconstructs a finite candidate set from Boundary Coordinates.

For a finite radix family  $B$ , the formulation is expressed as:

$$\begin{aligned} r\_b(n) &= ((n - 1) \bmod b), \\ D\_b(n) &= \min(r\_b(n), b - 1 - r\_b(n)), \\ A\_b(n) &= \{D\_b(n), b - 1 - D\_b(n)\}, \\ Cand\_B(n) &= \{x \in \mathbb{N} \mid \forall b \in B, (x - 1) \bmod b \in A\_b(n)\}, \\ \Phi\_B(n) &= Cand\_B(n) \cap [1, LIMIT] \cap \mathbb{P}, \end{aligned}$$

where  $\mathbb{P}$  denotes the set of prime numbers.

The reconstruction process consists of three consecutive stages:

- Boundary Coordinate generation.
- Candidate reconstruction from modular boundary constraints.
- Structural verification within the computational range.

Unlike exhaustive search, the proposed framework progressively reduces the candidate space through independent modular constraints. The mathematical framework itself remains unchanged throughout Verification001–030 and Applications001–004.

## 4. Verification001–030

### 4.1 Overview

The mathematical framework of IKERUSIKI Theory was established through a sequence of thirty computational verification stages. Rather than serving as independent experiments, these verification stages progressively constructed and validated the proposed structural representation of integers across five consecutive phases.

### 4.2 Phase I — Boundary Observation

Verification001–014 investigated modular boundary behavior for integers under multiple radix systems. These experiments demonstrated that Boundary Distance remained structurally stable across different bases and served as the fundamental observable quantity.

### 4.3 Phase II — Coordinate Construction

Verification015–024 introduced Boundary Coordinates as finite structural representations obtained from Boundary Distances, demonstrating consistency across multiple radix systems.

## 4.4 Phase III — Candidate Reconstruction

Verification025–028 established the Candidate Reconstruction framework, showing that Boundary Coordinates reconstruct finite candidate sets through independent modular constraints.

## 4.5 Phase IV — Uniqueness Verification

Verification029–030 evaluated structural uniqueness and practical reconstruction, introducing benchmark methodology and demonstrating structural reconstruction for RSA-oriented composite integers.

## 4.6 Summary

Verification001–030 collectively established the complete mathematical framework, forming the theoretical basis for the computational applications.

# 5. Applications

## 5.1 Application001: RSA2048 Structural Demonstration

### Objective

Application001 evaluates the completed mathematical framework on RSA2048-oriented composite integers, demonstrating that Boundary Coordinates preserve sufficient structural information for candidate reconstruction.

### Experimental Configuration

The experiment applies the mathematical framework established through Verification001–030 without modification, consisting of Boundary Coordinate generation, Candidate Reconstruction, and structural consistency verification.

### Experimental Results

Application001 confirmed that the completed IKERUSIKI mathematical framework successfully reconstructs structural candidates for RSA2048-oriented composite integers.

## 5.2 Application002: Large Integer Demonstration

### Objective

Application002 evaluates the scalability of the completed IKERUSIKI mathematical framework on large integers.

### Experimental Configuration

The evaluation used randomly generated integers with bit lengths of 512, 1024, 2048, and 4096 bits. Measured metrics include Candidate Count, Reduction Ratio, Reconstruction Consistency, Peak Memory Usage, and Execution Time.

**Table 1. Experimental Results for Large Integer Demonstration**

| Integer Size | Candidate Count                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Reduction Ratio   | Reconstruction | Peak Memory | Execution Time |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------|-------------|----------------|
| 512-bit      | 40066446856691257769<br>53793577298352961699<br>87139454296720396414<br>92333728591834343987<br>10815477667265327193<br>91406573061896780202<br>00746839382619952388<br>53861                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1673196525.000000 | True           | 8.270 KB    | 0.006547 sec   |
| 1024-bit     | 53720322388976868922<br>47475803205558096703<br>48490277406765691884<br>52371944973949040733<br>01631691989152545485<br>38821863378754606717<br>99682876108775274468<br>38542490980673187143<br>03131321843634824373<br>72263023942785039277<br>72384209492929653038<br>88443298319287129497<br>65667017854746773002<br>22886198019902232897<br>5885335655761157415                                                                                                                                                                                                                                                                                                                                                                                     | 1673196525.000000 | True           | 4.480 KB    | 0.060700 sec   |
| 2048-bit     | 96572654761254083111<br>11235629858229582578<br>20391711472268092666<br>08498139975335138503<br>50961373593434552902<br>77055708113005375130<br>51909903860489194761<br>76939940288330183040<br>17735373344625245088<br>82034612318199728682<br>91243931303370592470<br>79258061337470148473<br>39813147384410345497<br>32974446288542193673<br>59306720038532207802<br>97716438044638509120<br>56615909172649807101<br>85698484403537902460<br>62332115049459077117<br>99861738940157849319<br>03833493974761981963<br>29200951449355754255<br>00571250498387579811<br>41253778473055065071<br>02401366455524198878<br>75896270637024152896<br>51823693225546193332<br>41578320841412623492<br>03759980684159215941<br>02002003385352284181<br>1308167 | 1673196525.000000 | True           | 4.500 KB    | 0.044700 sec   |
| 4096-bit     | 31209390702420700601<br>55523902718553585867<br>51860797518623935772<br>72200106736056350507<br>20801690679864989434<br>94067472652215573672<br>07285272068856155652<br>82255435748545072868<br>70254689212314237970                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 836598262.500000  | True           | 8.227 KB    | 0.009481 sec   |

| Integer Size | Candidate Count                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Reduction Ratio | Reconstruction | Peak Memory | Execution Time |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------|-------------|----------------|
|              | 81760446033627592791<br>88712921652243977493<br>43699393813190095731<br>38594302826944010916<br>64362929726034569490<br>03228749374786617967<br>07622455973573602831<br>49601768703261531878<br>15340334708570119540<br>88986250328336625759<br>36716100542087294681<br>93772264703123109318<br>62529251741564969118<br>37276973475889510913<br>97153424167272184328<br>03891158289102739204<br>17894576878778828354<br>94154822970899017925<br>34146335071331540726<br>17061066296816359411<br>21232131516883488207<br>00380434827194725571<br>93821704867650653994<br>09698035507844135126<br>38561635857843089223<br>42540351604797529452<br>30513005722796108718<br>91301207404057332201<br>02502303522151393038<br>26497946525015724881<br>53469944448091145608<br>95751672282558885488<br>34664232790288075384<br>15001423185448835444<br>36023177091151679935<br>05894844712624183916<br>88347280242615736275<br>41177037934742096263<br>61724574398535092425<br>32757778326519650756<br>38667402800962671658<br>14493225310119340481<br>91086768156850944913<br>15554073385277386488<br>03297739725130290005<br>68845834084027903614<br>51745407337566274266<br>34717796518809211302<br>19539008607667978550<br>88820087017951083828<br>87512610511464211216<br>32243292229040599649<br>2374 |                 |                |             |                |

### 5.3 Application003: Computational Optimization

#### Objective

Application003 evaluates computational optimization while preserving the completed mathematical framework.

**Table 2. Computational Optimization Results**

| Integer Size | Baseline Time (sec) | Optimized Time (sec) | Baseline Memory (KB) | Optimized Memory (KB) | Reconstruction |
|--------------|---------------------|----------------------|----------------------|-----------------------|----------------|
| 512-bit      | 0.006547            | 0.007437             | 8.270                | 9.969                 | True           |
| 1024-bit     | 0.060700            | 0.013608             | 4.480                | 8.152                 | True           |
| 2048-bit     | 0.044700            | 0.011046             | 4.500                | 8.184                 | True           |
| 4096-bit     | 0.009481            | 0.008236             | 8.227                | 8.160                 | True           |

## 5.4 Application004: Benchmark Analysis

### Objective

Application004 compares the computational performance of the baseline and optimized implementations.

**Table 3. Benchmark Comparison**

| Metric          | Baseline   | Optimized  | Observation     |
|-----------------|------------|------------|-----------------|
| Candidate Count | Identical  | Identical  | No change       |
| Reduction Ratio | Identical  | Identical  | No change       |
| Reconstruction  | True       | True       | No change       |
| Peak Memory     | Comparable | Comparable | Nearly constant |
| Execution Time  | Baseline   | Reduced    | Improved        |

## 6. Discussion

### 6.1 Interpretation of the Framework

IKERUSIKI Theory represents integers through structural information derived from Boundary Coordinates rather than solely through their numerical values. The completed framework demonstrates that modular boundary information provides a consistent structural representation across multiple radix systems.

### 6.2 Computational Characteristics

As summarized in Table 2 and Table 3, implementation-level optimization reduced execution time while preserving identical structural reconstruction, reduction ratios, and reconstruction consistency.



## 6.3 Scope and Limitations

The present study establishes a computational framework for structural reconstruction based on Boundary Coordinates up to 4096 bits. The work evaluates structural reconstruction rather than proposing a practical factorization algorithm for cryptographic applications.

## 6.4 Future Work

Future work includes independent external verification, reproducibility studies by third parties, and further mathematical evaluation of structural representations derived from Boundary Coordinates.

## 7. Conclusion

This paper presented IKERUSIKI Theory as a structural mathematical framework for representing integers through Boundary Coordinates. The framework was established through Verification001–030 and evaluated through Applications001–004. Collectively, these results establish a unified computational framework in which mathematical structure remains invariant while implementation strategy evolves independently.

## References

- [1] R. Crandall and C. Pomerance, Prime Numbers: A Computational Perspective, Springer.
- [2] D. E. Knuth, The Art of Computer Programming, Vol. 2: Seminumerical Algorithms, Addison-Wesley.
- [3] T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, Introduction to Algorithms, MIT Press.
- [4] IKERUSIKI Theory GitHub Repository: <https://github.com/Hironori-Sushiki-S11/RSA-2048-Collapse-S11>

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## Appendix A: Verification Summary

- Verification001–014: Boundary Observation.
- Verification015–024: Boundary Coordinate Construction.
- Verification025–028: Candidate Reconstruction.
- Verification029: Benchmark Framework.
- Verification030: RSA-oriented Structural Reconstruction.

## Appendix B: Applications Summary

- Application001: RSA2048-oriented Structural Demonstration.
- Application002: Large Integer Demonstration (512–4096 bits).
- Application003: Computational Optimization.
- Application004: Benchmark Analysis.

## Appendix C: Repository Structure

- README.md
- IKERUSIKI\_THEORY.md
- IKERUSIKI\_Theory\_Final\_Edition.md
- verification001.py through verification030.py
- application002\_large\_integer.py
- application003\_computational\_optimization.py
- application004\_benchmark\_analysis.py

## Reproducibility

All computational experiments reported in this paper were performed using the public implementation contained in the IKERUSIKI Theory GitHub repository, ensuring complete reproducibility of the structural framework.